CYBERSRS

Secure Certificate Management System

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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Contents

1. Introduction 3

2. Functional requirements 4

3. Non-functional requirements 10

4. Security, privacy and compliance 12

5. Data requirements 12

6. Network and interfaces 12

7. Compliance and policy applicability matrix 13

8. Architecture 13

9. Threats 19

10. Risks 21

11. User stories and use cases 23

12. Verification and acceptance 25

13. Glossary 30

14. SDLC recommendation and workflow 30

SDLC rankings and decision evidence 35

Evidence register 37

Source references 37

Validation findings 38

Human review history 39

AI drafting record 39

Requirements traceability matrix 40

1. Introduction

Original project description

Design a secure digital-certificate management system for a university that manages certificates used by approved campus applications, servers, and network services. Authorized certificate administrators should be able to review certificate requests, approve certificate issuance, renew certificates, revoke compromised certificates, and track certificate status. The system should restrict certificate administration based on approved roles, monitor certificate validity, detect expired certificates and unauthorized certificate changes, generate security alerts, and maintain audit logs of certificate lifecycle activities and administrator actions. The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure digital-certificate management system for a university that manages certificates used by approved campus applications, servers, and network services.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: Authorized certificate administrators should be able to review certificate requests, approve certificate issuance, renew certificates, revoke compromised certificates, and track certificate status.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: The system should restrict certificate administration based on approved roles, monitor certificate validity, detect expired certificates and unauthorized certificate changes, generate security alerts, and maintain audit logs of certificate lifecycle activities and administrator actions.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

FR-001 - manage certificates used by approved campus applications, servers, and network services

Statement: The system shall manage certificates used by approved campus applications, servers, and network services.

Rationale: Source obligation E-001_O01.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: system manages certificates used by approved campus applications, servers, and network services

When: the system initiates certificate management for an approved service

Then: the resulting certificate state is returned in a response captured externally

Method: test

Observable evidence: External test capture of the resulting response

FR-002 - allow Authorized certificate administrators to review certificate requests

Statement: The system shall allow Authorized certificate administrators to review certificate requests.

Rationale: Source obligation E-002_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Authorized certificate administrators review a certificate request

When: the actor initiates the review operation on the named request

Then: the requested certificate request is returned for examination

Method: inspection

Observable evidence: External test capture of the resulting response

FR-003 - allow Authorized certificate administrators to approve certificate issuance

Statement: The system shall allow Authorized certificate administrators to approve certificate issuance.

Rationale: Source obligation E-002_O02.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Authorized certificate administrators approve a certificate issuance request

When: the actor initiates the approval operation on the named request

Then: the request's approved status is observed in the system state

Method: demonstration

Observable evidence: External test capture of the resulting state

FR-004 - allow Authorized certificate administrators to renew certificates

Statement: The system shall allow Authorized certificate administrators to renew certificates.

Rationale: Source obligation E-002_O03.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Authorized certificate administrators renew a certificate

When: the actor initiates the renewal operation on the named certificate

Then: the certificate is observed in its renewed state

Method: analysis

Observable evidence: External test capture of the resulting state

FR-005 - allow Authorized certificate administrators to revoke compromised certificates

Statement: The system shall allow Authorized certificate administrators to revoke compromised certificates.

Rationale: Source obligation E-002_O04.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Authorized certificate administrators revoke a compromised certificate

When: the actor initiates the revocation operation on the named certificate

Then: the certificate's revoked status is observed in the system state

Method: test

Observable evidence: External test capture of the resulting state

FR-006 - allow Authorized certificate administrators to track certificate status

Statement: The system shall allow Authorized certificate administrators to track certificate status.

Rationale: Source obligation E-002_O05.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Authorized certificate administrators track a certificate status

When: the actor initiates the tracking operation on the named certificate

Then: the history of that certificate's status is available for examination

Method: inspection

Observable evidence: External test capture of the resulting state

FR-007 - restrict certificate administration based on approved roles

Statement: The system shall restrict certificate administration based on approved roles.

Rationale: Source obligation E-003_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: the system has certificate administration privileges assigned to approved roles

When: the system attempts to perform certificate administration actions

Then: only actions permitted by the approved role are executed; unauthorized actions are denied

Method: test

Observable evidence: External test capture of the resulting response

FR-008 - monitor certificate validity

Statement: The system shall monitor certificate validity.

Rationale: Source obligation E-003_O02.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: a certificate is issued with a defined validity period

When: the system monitors certificate validity during operation

Then: The system reports that certificate validity is within the defined period during operation

Method: inspection

Observable evidence: External test capture of the resulting state

FR-009 - detect expired certificates

Statement: The system shall detect expired certificates.

Rationale: Source obligation E-003_O03.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: a certificate has expired or is nearing expiration

When: the system detects certificate validity status changes

Then: the system identifies and flags expired certificates within the monitoring scope

Method: analysis

Observable evidence: External test capture of the inspection record

FR-010 - detect unauthorized certificate changes

Statement: The system shall detect unauthorized certificate changes.

Rationale: Source obligation E-003_O04.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: a certificate is modified by an unauthorized entity

When: An unauthorized modification to the certificate is detected.

Then: The system generates an alert indicating the change and its source.

Method: demonstration

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-dad9a79bf709 is TBD.

FR-011 - generate security alerts

Statement: The system shall generate security alerts.

Rationale: Source obligation E-003_O05.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: a security condition is triggered that requires alerting

When: the system initiates an alert condition

Then: an alert exists in the system as a record, with no specified destination or field set

Method: test

Observable evidence: External test capture of the resulting response

FR-012 - maintain audit logs of certificate lifecycle activities and administrator actions

Statement: The system shall maintain audit logs of certificate lifecycle activities and administrator actions.

Rationale: Source obligation E-003_O06.

Priority: should

Dimensions: business; auditability; data

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: certificate lifecycle activities and administrator actions occur

When: the system records all administrative interactions with certificates

Then: The system records all administrative interactions with certificates in audit logs

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-3aae42e94793 is TBD.; Decision Q-64ef2e28335c is TBD.

FR-013 - provide a centralized dashboard

Statement: The system shall provide a centralized dashboard.

Rationale: Source obligation E-004_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: a system-controlled user opens the dashboard named in the source obligation

When: the system provides a centralized dashboard as specified

Then: the requested dashboard is displayed or available for use

Method: test

Observable evidence: External test capture of the resulting response

FR-014 - integrate with existing university identity services

Statement: The system shall integrate with existing university identity services.

Rationale: Source obligation E-004_O02.

Priority: should

Dimensions: business; integration

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: a system-controlled user initiates an exchange with the named external identity service

When: the system integrates with existing university identity services as specified

Then: interoperation with that service is observed

Method: test

Observable evidence: External test capture of the resulting response

FR-015 - support multiple campus locations

Statement: The system shall support multiple campus locations.

Rationale: Source obligation E-004_O03.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: a user initiates use from each stated campus location

When: the system supports multiple campus locations as specified

Then: use succeeds at each location

Method: test

Observable evidence: External test capture of the resulting state

3. Non-functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

NFR-001 - be secure

Statement: The system shall be secure.

Rationale: Source obligation E-001_O02.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data leakage via certificate management operations.

When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD.

Then: The plan's measurable targets remain TBD or unconfirmed, as no specific thresholds, attack scenarios, or success criteria are provided in the current draft.

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-002 - remain secure

Statement: The system shall remain secure.

Rationale: Source obligation E-004_O04.

Priority: should

Dimensions: security

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain secure against unauthorized access and data breaches.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-003 - remain reliable

Statement: The system shall remain reliable.

Rationale: Source obligation E-004_O05.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain reliable under varying loads and failure conditions.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-819bb986f419 is TBD.

NFR-004 - remain scalable

Statement: The system shall remain scalable.

Rationale: Source obligation E-004_O06.

Priority: should

Dimensions: performance

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain scalable to support increasing numbers of users and data volumes.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the resulting state

Unresolved decisions: Decision Q-543394346815 is TBD.

NFR-005 - remain manageable

Statement: The system shall remain manageable.

Rationale: Source obligation E-004_O07.

Priority: should

Dimensions: usability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain manageable through configuration, monitoring, and operational controls.

When: An analyst inspects or reviews that acceptance plan.

Then: The plan's measurable targets remain TBD or unconfirmed.

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-5268b96e41b0 is TBD.

Related requirement ids: FR-001; FR-013; FR-014; FR-015

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; NFR-001; NFR-002

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-007; FR-008; FR-009; FR-010; FR-011; FR-012

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001

7. Compliance and policy applicability matrix

Open decisions: Decision Q-66cf5513b374 is TBD.; Decision Q-dad9a79bf709 is TBD.

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: Authorized certificate administrators

Detail: Proposed logical endpoint for the source-stated actor: Authorized certificate administrators.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

Component name: Actor: Authorized certificate administrators

Responsibility: Represent the source-stated actor: Authorized certificate administrators.

Certificate Administration Module

Detail: Proposed logical owner for FR-002, FR-003, FR-004, FR-005, FR-006, FR-001, FR-007, NFR-001, NFR-003.

Status: proposed

Evidence ids: E-001; E-002; E-003; E-004

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-001; FR-007; NFR-001; NFR-003

Component name: Certificate Administration Module

Responsibility: The system shall allow Authorized certificate administrators to review certificate requests. The system shall allow Authorized certificate administrators to approve certificate issuance. The system shall allow Authorized certificate administrators to renew certificates. The system shall allow Authorized certificate administrators to revoke compromised certificates. The system shall allow Authorized certificate administrators to track certificate status. The system shall manage certificates used by approved campus applications, servers, and network services. The system shall restrict certificate administration based on approved roles. The system shall be secure. The system shall remain reliable.

Certificate Monitoring and Alerting Module

Detail: Proposed logical owner for FR-008, FR-009, FR-010, FR-011, FR-001, NFR-002, NFR-003.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-008; FR-009; FR-010; FR-011; FR-001; NFR-002; NFR-003

Component name: Certificate Monitoring and Alerting Module

Responsibility: The system shall monitor certificate validity. The system shall detect expired certificates. The system shall detect unauthorized certificate changes. The system shall generate

security alerts. The system shall manage certificates used by approved campus applications, servers, and network services. The system shall remain secure. The system shall remain reliable.

Audit and Logging Module

Detail: Proposed logical owner for FR-012, FR-001, NFR-001, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-012; FR-001; NFR-001; NFR-005

Component name: Audit and Logging Module

Responsibility: The system shall maintain audit logs of certificate lifecycle activities and administrator actions. The system shall manage certificates used by approved campus applications, servers, and network services. The system shall be secure. The system shall remain manageable.

Centralized Dashboard Module

Detail: Proposed logical owner for FR-013, FR-015, NFR-002, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-004

Requirement ids: FR-013; FR-015; NFR-002; NFR-004; NFR-005

Component name: Centralized Dashboard Module

Responsibility: The system shall provide a centralized dashboard. The system shall support multiple campus locations. The system shall remain secure. The system shall remain scalable. The system shall remain manageable.

Integration with Identity Services Module

Detail: Proposed logical owner for FR-014, FR-007, FR-015, NFR-004.

Status: proposed

Evidence ids: E-003; E-004

Requirement ids: FR-014; FR-007; FR-015; NFR-004

Component name: Integration with Identity Services Module

Responsibility: The system shall integrate with existing university identity services. The system shall restrict certificate administration based on approved roles. The system shall support multiple campus locations. The system shall remain scalable.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-002, FR-003, FR-004, FR-005, FR-006.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

From component: Actor: Authorized certificate administrators

To component: Certificate Administration Module

Data or action: The system shall allow Authorized certificate administrators to review certificate requests. The system shall allow Authorized certificate administrators to approve certificate issuance. The system shall allow Authorized certificate administrators to renew certificates. The system shall allow Authorized certificate administrators to revoke compromised certificates. The system shall allow Authorized certificate administrators to track certificate status.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, NFR-003.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-001; NFR-003

From component: Certificate Administration Module

To component: Certificate Monitoring and Alerting Module

Data or action: The system shall manage certificates used by approved campus applications, servers, and network services. The system shall remain reliable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: FR-001; NFR-001

From component: Certificate Administration Module

To component: Audit and Logging Module

Data or action: The system shall manage certificates used by approved campus applications, servers, and network services. The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for FR-007.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-007

From component: Certificate Administration Module

To component: Integration with Identity Services Module

Data or action: The system shall restrict certificate administration based on approved roles.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, NFR-003.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: FR-001; NFR-003

From component: Certificate Monitoring and Alerting Module

To component: Certificate Administration Module

Data or action: The system shall manage certificates used by approved campus applications, servers, and network services. The system shall remain reliable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: FR-001

From component: Certificate Monitoring and Alerting Module

To component: Audit and Logging Module

Data or action: The system shall manage certificates used by approved campus applications, servers, and network services.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-002.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-002

From component: Certificate Monitoring and Alerting Module

To component: Centralized Dashboard Module

Data or action: The system shall remain secure.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001, NFR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: FR-001; NFR-001

From component: Audit and Logging Module

To component: Certificate Administration Module

Data or action: The system shall manage certificates used by approved campus applications, servers, and network services. The system shall be secure.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-001

Requirement ids: FR-001

From component: Audit and Logging Module

To component: Certificate Monitoring and Alerting Module

Data or action: The system shall manage certificates used by approved campus applications, servers, and network services.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-005.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-005

From component: Audit and Logging Module

To component: Centralized Dashboard Module

Data or action: The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-002.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-002

From component: Centralized Dashboard Module

To component: Certificate Monitoring and Alerting Module

Data or action: The system shall remain secure.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-005.

Status: proposed

Evidence ids: E-004

Requirement ids: NFR-005

From component: Centralized Dashboard Module

To component: Audit and Logging Module

Data or action: The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-015, NFR-004.

Status: proposed

Evidence ids: E-004

Requirement ids: FR-015; NFR-004

From component: Centralized Dashboard Module

To component: Integration with Identity Services Module

Data or action: The system shall support multiple campus locations. The system shall remain scalable.

Proposed source-defined interaction

Detail: Proposed directed path for FR-007.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-007

From component: Integration with Identity Services Module

To component: Certificate Administration Module

Data or action: The system shall restrict certificate administration based on approved roles.

Proposed source-defined interaction

Detail: Proposed directed path for FR-015, NFR-004.

Status: proposed

Evidence ids: E-004

Requirement ids: FR-015; NFR-004

From component: Integration with Identity Services Module

To component: Centralized Dashboard Module

Data or action: The system shall support multiple campus locations. The system shall remain scalable.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002, FR-003, FR-004, FR-005, FR-006.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

Untrusted component: Actor: Authorized certificate administrators

Trusted component: Certificate Administration Module

Enforcement: The system shall allow Authorized certificate administrators to review certificate requests. The system shall allow Authorized certificate administrators to approve certificate issuance. The system shall allow Authorized certificate administrators to renew certificates. The system shall allow Authorized certificate administrators to revoke compromised certificates. The system shall allow Authorized certificate administrators to track certificate status.

9. Threats

Threats

Id: THR-001

Asset: User identities and authenticated access

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Impact: Unauthorized use of a legitimate identity could expose resources or alter system state.

Control requirement ids: FR-013; FR-014; FR-015; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-002

Asset: Role and permission boundaries

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Impact: A lower-privilege user could access restricted functions or sensitive resources.

Control requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012

Evidence ids: E-002; E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-003

Asset: Access grants and revocations

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: A revoked or changed permission remains effective in a session or enforcement point, allowing access after authorization should have ended.

Impact: A formerly authorized principal could continue using access that is no longer approved.

Control requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

Evidence ids: E-002

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-004

Asset: Integration trust paths

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Impact: Untrusted integration data could cause incorrect decisions or hide security events.

Control requirement ids: FR-013; FR-014; FR-015; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-005

Asset: Security and administrative audit evidence

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Impact: Investigators could lose reliable evidence of the events required by the brief.

Control requirement ids: FR-007; FR-008; FR-009; FR-010; FR-011; FR-012

Evidence ids: E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-006

Asset: Monitoring and security alerts

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Impact: Relevant suspicious activity could remain undetected or receive delayed investigation.

Control requirement ids: FR-007; FR-008; FR-009; FR-010; FR-011; FR-012

Evidence ids: E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

Risks

Id: RISK-001

Description: Derived risk affecting User identities and authenticated access: Unauthorized use of a legitimate identity could expose resources or alter system state.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Mitigation: Review impersonation and compromised-identity paths against the stated access duties; confirm any additional identity or session protections with the stakeholder. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-013; FR-014; FR-015; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Owner: TBD

Id: RISK-002

Description: Derived risk affecting Role and permission boundaries: A lower-privilege user could access restricted functions or sensitive resources.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Mitigation: Test permitted and denied role/resource pairs against the linked authorization duties; review ambiguous permission assignments with the resource owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012

Evidence ids: E-002; E-003

Owner: TBD

Id: RISK-003

Description: Derived risk affecting Access grants and revocations: A formerly authorized principal could continue using access that is no longer approved.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A revoked or changed permission remains effective in a session or enforcement point, allowing access after authorization should have ended.

Mitigation: Trace permission changes through subsequent access decisions and any active access; confirm revocation timing and session semantics before baselining them. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006

Evidence ids: E-002

Owner: TBD

Id: RISK-004

Description: Derived risk affecting Integration trust paths: Untrusted integration data could cause incorrect decisions or hide security events.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Mitigation: Review trust assumptions at the specified integration boundaries with missing or altered input fixtures; confirm the required failure handling with the owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-013; FR-014; FR-015; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Owner: TBD

Id: RISK-005

Description: Derived risk affecting Security and administrative audit evidence: Investigators could lose reliable evidence of the events required by the brief.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Mitigation: Compare source-required events with the resulting audit evidence; assess missing and altered-record cases and confirm unprovided protection or retention decisions. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-007; FR-008; FR-009; FR-010; FR-011; FR-012

Evidence ids: E-003

Owner: TBD

Id: RISK-006

Description: Derived risk affecting Monitoring and security alerts: Relevant suspicious activity could remain undetected or receive delayed investigation.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Mitigation: Evaluate the stated detection duties with representative positive and negative events; confirm undecided thresholds, alert priorities and investigation responsibilities. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-007; FR-008; FR-009; FR-010; FR-011; FR-012

Evidence ids: E-003

Owner: TBD

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: Authorized certificate administrators

Goal: The system shall allow Authorized certificate administrators to review certificate requests.

Story: As Authorized certificate administrators, I need the following behavior: The system shall allow Authorized certificate administrators to review certificate requests.

Preconditions

1. Authorized certificate administrators review a certificate request

Steps

1. the actor initiates the review operation on the named request

Success outcome: the requested certificate request is returned for examination

Requirement ids: FR-002

Evidence ids: E-002

Id: UC-002

Actor: Authorized certificate administrators

Goal: The system shall allow Authorized certificate administrators to approve certificate issuance.

Story: As Authorized certificate administrators, I need the following behavior: The system shall allow Authorized certificate administrators to approve certificate issuance.

Preconditions

1. Authorized certificate administrators approve a certificate issuance request

Steps

1. the actor initiates the approval operation on the named request

Success outcome: the request's approved status is observed in the system state

Requirement ids: FR-003

Evidence ids: E-002

Id: UC-003

Actor: Authorized certificate administrators

Goal: The system shall allow Authorized certificate administrators to renew certificates.

Story: As Authorized certificate administrators, I need the following behavior: The system shall allow Authorized certificate administrators to renew certificates.

Preconditions

1. Authorized certificate administrators renew a certificate

Steps

1. the actor initiates the renewal operation on the named certificate

Success outcome: the certificate is observed in its renewed state

Requirement ids: FR-004

Evidence ids: E-002

Id: UC-004

Actor: Authorized certificate administrators

Goal: The system shall allow Authorized certificate administrators to revoke compromised certificates.

Story: As Authorized certificate administrators, I need the following behavior: The system shall allow Authorized certificate administrators to revoke compromised certificates.

Preconditions

1. Authorized certificate administrators revoke a compromised certificate

Steps

1. the actor initiates the revocation operation on the named certificate

Success outcome: the certificate's revoked status is observed in the system state

Requirement ids: FR-005

Evidence ids: E-002

Id: UC-005

Actor: Authorized certificate administrators

Goal: The system shall allow Authorized certificate administrators to track certificate status.

Story: As Authorized certificate administrators, I need the following behavior: The system shall allow Authorized certificate administrators to track certificate status.

Preconditions

1. Authorized certificate administrators track a certificate status

Steps

1. the actor initiates the tracking operation on the named certificate

Success outcome: the history of that certificate's status is available for examination

Requirement ids: FR-006

Evidence ids: E-002

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: system manages certificates used by approved campus applications, servers, and network services
2. When: the system initiates certificate management for an approved service

Expected result: Expected: the resulting certificate state is returned in a response captured externally

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: Authorized certificate administrators review a certificate request
2. When: the actor initiates the review operation on the named request

Expected result: Expected: the requested certificate request is returned for examination

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: Authorized certificate administrators approve a certificate issuance request
2. When: the actor initiates the approval operation on the named request

Expected result: Expected: the request's approved status is observed in the system state

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-004

Requirement ids: FR-004

Procedure

1. Given: Authorized certificate administrators renew a certificate
2. When: the actor initiates the renewal operation on the named certificate

Expected result: Expected: the certificate is observed in its renewed state

Evidence to record: Evidence: External test capture of the resulting state

Test type: analysis

Id: TEST-005

Requirement ids: FR-005

Procedure

1. Given: Authorized certificate administrators revoke a compromised certificate
2. When: the actor initiates the revocation operation on the named certificate

Expected result: Expected: the certificate's revoked status is observed in the system state

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-006

Requirement ids: FR-006

Procedure

1. Given: Authorized certificate administrators track a certificate status
2. When: the actor initiates the tracking operation on the named certificate

Expected result: Expected: the history of that certificate's status is available for examination

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-007

Requirement ids: FR-007

Procedure

1. Given: the system has certificate administration privileges assigned to approved roles
2. When: the system attempts to perform certificate administration actions

Expected result: Expected: only actions permitted by the approved role are executed; unauthorized actions are denied

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-008

Requirement ids: FR-008

Procedure

1. Given: a certificate is issued with a defined validity period
2. When: the system monitors certificate validity during operation

Expected result: Expected: The system reports that certificate validity is within the defined period during operation

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-009

Requirement ids: FR-009

Procedure

1. Given: a certificate has expired or is nearing expiration
2. When: the system detects certificate validity status changes

Expected result: Expected: the system identifies and flags expired certificates within the monitoring scope

Evidence to record: Evidence: External test capture of the inspection record

Test type: analysis

Id: TEST-010

Requirement ids: FR-010

Procedure

1. Given: a certificate is modified by an unauthorized entity
2. When: An unauthorized modification to the certificate is detected.

Expected result: Expected: The system generates an alert indicating the change and its source.

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-011

Requirement ids: FR-011

Procedure

1. Given: a security condition is triggered that requires alerting
2. When: the system initiates an alert condition

Expected result: Expected: an alert exists in the system as a record, with no specified destination or field set

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-012

Requirement ids: FR-012

Procedure

1. Given: certificate lifecycle activities and administrator actions occur
2. When: the system records all administrative interactions with certificates

Expected result: Expected: The system records all administrative interactions with certificates in audit logs

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-013

Requirement ids: FR-013

Procedure

1. Given: a system-controlled user opens the dashboard named in the source obligation
2. When: the system provides a centralized dashboard as specified

Expected result: Expected: the requested dashboard is displayed or available for use

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-014

Requirement ids: FR-014

Procedure

1. Given: a system-controlled user initiates an exchange with the named external identity service
2. When: the system integrates with existing university identity services as specified

Expected result: Expected: interoperation with that service is observed

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-015

Requirement ids: FR-015

Procedure

1. Given: a user initiates use from each stated campus location
2. When: the system supports multiple campus locations as specified

Expected result: Expected: use succeeds at each location

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-016

Requirement ids: NFR-001

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data leakage via certificate management operations.
2. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed, as no specific thresholds, attack scenarios, or success criteria are provided in the current draft.

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-017

Requirement ids: NFR-002

Procedure

1. Given: The system shall remain secure against unauthorized access and data breaches.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-018

Requirement ids: NFR-003

Procedure

1. Given: The system shall remain reliable under varying loads and failure conditions.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-019

Requirement ids: NFR-004

Procedure

1. Given: The system shall remain scalable to support increasing numbers of users and data volumes.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-020

Requirement ids: NFR-005

Procedure

1. Given: The system shall remain manageable through configuration, monitoring, and operational controls.
2. When: An analyst inspects or reviews that acceptance plan.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed.

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-003

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-003

14. SDLC recommendation and workflow

Selected model: Incremental

Justification: The Incremental model is selected based on derived evidence showing the need for integrated security review (E-002), operational validation through monitoring feedback (E-003), staged deployment across multiple locations (E-004) to enable controlled rollout, and integration verification of connected systems (E-004). These factors collectively support a phased, iterative approach that aligns with continuous security assurance and operational feedback. The Incremental model scores 66 points—outperforming Hybrid (65), DevSecOps (63), Spiral (51), and others—due to superior alignment across derived validation criteria. While the evidence is derived rather than direct, it establishes a consistent need for iterative validation, security integration, and deployment control. The model does not require assumptions about budget, delivery, failure consequences, or legacy systems; these remain unknown factors as specified. The decision remains proposed, pending confirmation of team capacity and delivery constraints. No product-based suitability is inferred beyond the evidence provided.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
2. Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall manage certificates used by approved campus applications, servers, and network services.; FR-002: The system shall allow Authorized certificate administrators to review certificate requests.; FR-003: The system shall allow Authorized certificate administrators to approve certificate issuance.; FR-004: The system shall allow Authorized certificate administrators to renew certificates.; FR-005: The system shall allow Authorized certificate administrators to revoke compromised certificates.; FR-006: The system shall allow Authorized certificate administrators to track certificate status.; FR-007: The system shall restrict certificate administration based on approved roles.; FR-008: The system shall monitor certificate validity.; FR-009: The system shall detect expired certificates.; FR-010: The system shall detect unauthorized certificate changes.; FR-011: The system shall generate security alerts.; FR-012: The system shall maintain audit logs of certificate lifecycle activities and administrator actions.; FR-013: The system shall provide a centralized dashboard.; FR-014: The system shall integrate with existing university identity services.; FR-015: The system shall support multiple campus locations.; NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
2. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against FR-011: The system shall generate security alerts.
2. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
3. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; FR-013; FR-014; FR-015; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Name: Incremental delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.

2. Implement the source-defined work group: FR-001: The system shall manage certificates used by approved campus applications, servers, and network services.; FR-002: The system shall allow Authorized certificate administrators to review certificate requests.; FR-003: The system shall allow Authorized certificate administrators to approve certificate issuance.; FR-004: The system shall allow Authorized certificate administrators to renew certificates.; FR-005: The system shall allow Authorized certificate administrators to revoke compromised certificates.; FR-006: The system shall allow Authorized certificate administrators to track certificate status.; FR-007: The system shall restrict certificate administration based on approved roles.; FR-008: The system shall monitor certificate validity.; FR-009: The system shall detect expired certificates.; FR-010: The system shall detect unauthorized certificate changes.; FR-011: The system shall generate security alerts.; FR-012: The system shall maintain audit logs of certificate lifecycle activities and administrator actions.; FR-013: The system shall provide a centralized dashboard.; FR-014: The system shall integrate with existing university identity services.; FR-015: The system shall support multiple campus locations.; NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
3. In an authorized isolated test environment, execute TEST-001: Given: system manages certificates used by approved campus applications, servers, and network services When: the system initiates certificate management for an approved service Expected: the resulting certificate state is returned in a response captured externally
4. In an authorized isolated test environment, execute TEST-002: Given: Authorized certificate administrators review a certificate request When: the actor initiates the review operation on the named request Expected: the requested certificate request is returned for examination
5. In an authorized isolated test environment, execute TEST-003: Given: Authorized certificate administrators approve a certificate issuance request When: the actor initiates the approval operation on the named request Expected: the request's approved status is observed in the system state
6. In an authorized isolated test environment, execute TEST-004: Given: Authorized certificate administrators renew a certificate When: the actor initiates the renewal operation on the named certificate Expected: the certificate is observed in its renewed state
7. In an authorized isolated test environment, execute TEST-005: Given: Authorized certificate administrators revoke a compromised certificate When: the actor initiates the revocation operation on the named certificate Expected: the certificate's revoked status is observed in the system state
8. In an authorized isolated test environment, execute TEST-006: Given: Authorized certificate administrators track a certificate status When: the actor initiates the tracking operation on the named certificate Expected: the history of that certificate's status is available for examination
9. In an authorized isolated test environment, execute TEST-007: Given: the system has certificate administration privileges assigned to approved roles When: the system attempts to perform certificate administration actions Expected: only actions permitted by the approved role are executed; unauthorized actions are denied
10. In an authorized isolated test environment, execute TEST-008: Given: a certificate is issued with a defined validity period When: the system monitors certificate validity during operation Expected: The system reports that certificate validity is within the defined period during operation
11. In an authorized isolated test environment, execute TEST-009: Given: a certificate has expired or is nearing expiration When: the system detects certificate validity status changes Expected: the system identifies and flags expired certificates within the monitoring scope
12. In an authorized isolated test environment, execute TEST-010: Given: a certificate is modified by an unauthorized entity When: An unauthorized modification to the certificate is detected. Expected: The system generates an alert indicating the change and its source.

13. In an authorized isolated test environment, execute TEST-011: Given: a security condition is triggered that requires alerting When: the system initiates an alert condition Expected: an alert exists in the system as a record, with no specified destination or field set
14. In an authorized isolated test environment, execute TEST-012: Given: certificate lifecycle activities and administrator actions occur When: the system records all administrative interactions with certificates Expected: The system records all administrative interactions with certificates in audit logs
15. In an authorized isolated test environment, execute TEST-013: Given: a system-controlled user opens the dashboard named in the source obligation When: the system provides a centralized dashboard as specified Expected: the requested dashboard is displayed or available for use
16. In an authorized isolated test environment, execute TEST-014: Given: a system-controlled user initiates an exchange with the named external identity service When: the system integrates with existing university identity services as specified Expected: interoperation with that service is observed
17. In an authorized isolated test environment, execute TEST-015: Given: a user initiates use from each stated campus location When: the system supports multiple campus locations as specified Expected: use succeeds at each location
18. In an authorized isolated test environment, execute TEST-016: Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data leakage via certificate management operations. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD. Expected: The plan's measurable targets remain TBD or unconfirmed, as no specific thresholds, attack scenarios, or success criteria are provided in the current draft.
19. In an authorized isolated test environment, execute TEST-017: Given: The system shall remain secure against unauthorized access and data breaches. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
20. In an authorized isolated test environment, execute TEST-018: Given: The system shall remain reliable under varying loads and failure conditions. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
21. In an authorized isolated test environment, execute TEST-019: Given: The system shall remain scalable to support increasing numbers of users and data volumes. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
22. In an authorized isolated test environment, execute TEST-020: Given: The system shall remain manageable through configuration, monitoring, and operational controls. When: An analyst inspects or reviews that acceptance plan. Expected: The plan's measurable targets remain TBD or unconfirmed.
23. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
24. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against FR-011: The system shall generate security alerts.
2. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
3. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; FR-013; FR-014; FR-015; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against FR-011: The system shall generate security alerts.
2. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
3. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; FR-013; FR-014; FR-015; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-2.0

Generation profile: sequential-sdlc-generation-2.1

Factors

Name: security_assurance

Value: integrated security review

Evidence ids: E-002

Basis: derived

Rationale: Security decisions benefit from security review and tests throughout delivery.

Name: operational_validation

Value: monitoring feedback

Evidence ids: E-003

Basis: derived

Rationale: Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency.

Name: deployment_scope

Value: staged rollout opportunity

Evidence ids: E-004

Basis: derived

Rationale: Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed.

Name: integration_assurance

Value: integration validation

Evidence ids: E-004

Basis: derived

Rationale: Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified.

Ranked candidates

Model: Incremental

Score: 66

Reasons: Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003); Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004); Derived

integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: Hybrid

Score: 65

Reasons: Derived security_assurance (+8 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived operational_validation (+5 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003); Derived deployment_scope (+8 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004); Derived integration_assurance (+4 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: DevSecOps

Score: 63

Reasons: Derived security_assurance (+12 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived operational_validation (+8 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003); Derived deployment_scope (+3 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)

Model: Spiral

Score: 51

Reasons: Derived security_assurance (+5 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived integration_assurance (+6 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure digital-certificate management system for a university that manages certificates used by approved campus applications, servers, and network services.

Id: E-002

Source: description

Quote: Authorized certificate administrators should be able to review certificate requests, approve certificate issuance, renew certificates, revoke compromised certificates, and track certificate status.

Id: E-003

Source: description

Quote: The system should restrict certificate administration based on approved roles, monitor certificate validity, detect expired certificates and unauthorized certificate changes, generate security alerts, and maintain audit logs of certificate lifecycle activities and administrator actions.

Id: E-004

Source: description

Quote: The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

Source references

kb-92ffa375cd05a8f5:12 - 03_OWASP_CheatSheets/081_OWASP_Multifactor_Authentication_Cheat_Sheet.md

Source document id: kb-92ffa375cd05a8f5

Chunk index: 12

Page or section: text chunk 13

Relevance score: 0.7592

Sha256: 0c1de91561bd8b449f5a27033e799c6928df3f24cb48a965994f615671612490

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7171

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-0150d5352d0ec304:8 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7131

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.6941

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-0150d5352d0ec304:1 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7169

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: qwen3:4b-instruct-2507-q4_K_M

Prompt version: sequential-1.38

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-002	E-002	UC-001, TEST-002, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-003	E-002	UC-002, TEST-003, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-004	E-002	UC-003, TEST-004, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-005	E-002	UC-004, TEST-005, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-006	E-002	UC-005, TEST-006, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-007	E-003	TEST-007, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-008	E-003	TEST-008, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-009	E-003	TEST-009, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-010	E-003	TEST-010, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-011	E-003	TEST-011, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-012	E-003	TEST-012, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-013	E-004	TEST-013, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-014	E-004	TEST-014, Requirements and verification planning, Incremental delivery cycle, Release and handoff

Requirement	Evidence / sources	Stories / tests / SDLC
FR-015	E-004	TEST-015, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-001	E-001	TEST-016, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-002	E-004	TEST-017, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-003	E-004	TEST-018, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-004	E-004	TEST-019, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-005	E-004	TEST-020, Requirements and verification planning, Incremental delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.38